



1. For the questions 1-6 consider the following and choose the correct answer:
 $v(t) = 200 \cos(628t + 60^\circ)$ and $i(t) = -20 \sin(628t - 130^\circ)$ 20L-40

179.908

- the voltage leads the current in degrees by:
- 1) ☒ a) 100 ☐ b) 628 ☐ c) 314 ☐ d) _____
 - 2) The current lags the voltage in radians by:
☐ a) 1.57 ☐ b) 3.14 ☒ c) 1.74 ☐ d) _____
 - 3) The voltage value (v) at $t = 5\text{ms}$ is:
☐ a) 50 V ☒ b) -100 V ☐ c) -150 V ☐ d) _____
 - 4) The phasor domain representation of $i(t)$ is:
☐ a) $20 \angle -130^\circ$ ☐ b) $40 \angle -40^\circ$ ☒ c) $20 \angle -40^\circ$ ☐ d) _____
 - 5) The phasor domain representation of $v(t)$ is:
☒ a) $200 \angle 60^\circ$ ☐ b) $100 \angle -60^\circ$ ☐ c) $200 \angle 130^\circ$ ☐ d) _____
 - 6) The frequency of the voltage in Hz is:
☐ a) 50 ☒ b) 100 ☐ c) 314 ☐ d) _____

Problem 2. For the questions 7-10 consider the circuit shown in figure 1 and answer using nodal analysis:

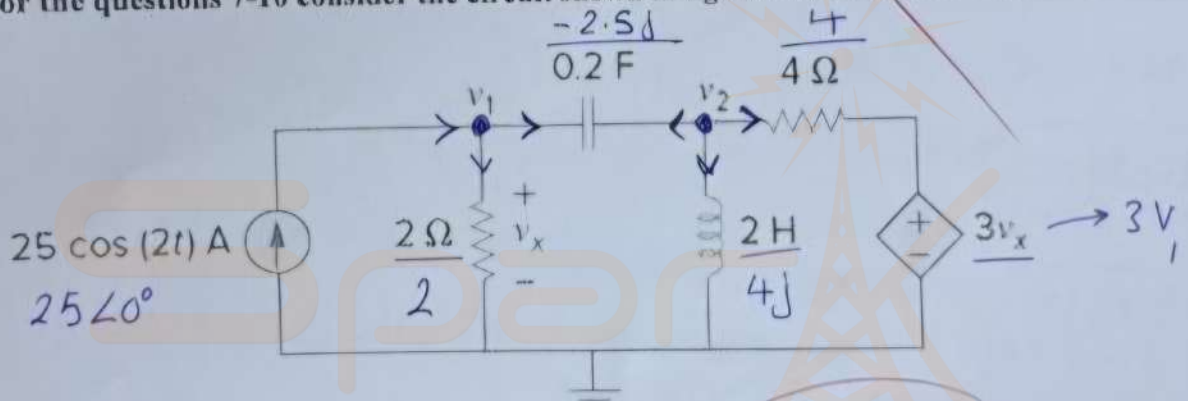


Figure 1

- 7) The voltage (V_x) is:
☒ a) $28.31 \angle 60^\circ$ ☐ b) $67.6 + j10$ V ☐ c) $77.682 \angle 50.19^\circ$ ☐ d) _____
- 8) The current through the inductor is:
☒ a) $24 \angle -90^\circ$ A ☐ b) $15 + j10$ A ☐ c) $20.64 \angle -32.9^\circ$ A ☐ d) _____
- 9) The voltage of node 2 with respect to node 1 is:
☒ a) $12 \angle 0^\circ$ V ☐ b) $5 + j10$ V ☐ c) $54.3 \angle 55.6^\circ$ V ☐ d) _____
- 10) write the required nodal equations (do not simplify):

① node (1) $\rightarrow 25 \angle 0^\circ = \frac{V_1}{2} + \frac{V_1 - V_2}{-2.5j}$

② node (2) $\rightarrow \frac{V_2 - V_1}{-2.5j} + \frac{V_2}{4j} + \frac{V_2 - 3V_1}{4} = 0$

without simplify

$\sum I_{in} = \sum I_{out}$

$V_x = V_1$

so $3V_x = 3V_1$



3. For the questions 11-16 consider the circuit shown in figure 2 and answer:

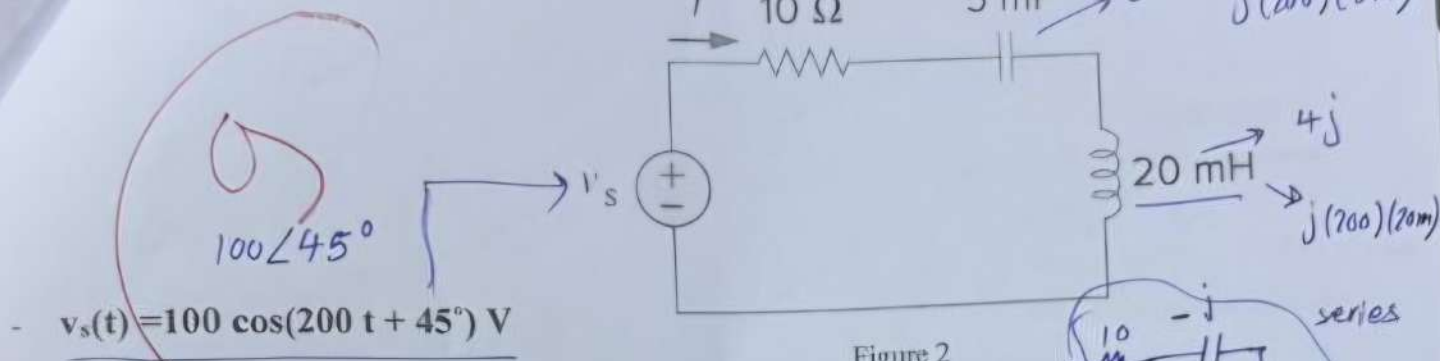


Figure 2

11) The equivalent impedance at the terminals of the voltage source in (Ω) is:

$$Z_{th} = 10 + j3 = 10.44 \angle 16.69^\circ \Omega$$

12) The admittance at the terminals of the voltage source in (S) is:

$$Y = \frac{1}{Z} = \frac{1}{10 + j3} = 0.0917 - j0.0275 \text{ S}$$

13) The steady state current (i) in time domain is:

$$I = \frac{V}{Z} = \frac{100 \angle 45^\circ}{10 + j3} = 9.57 \angle 28.3^\circ = 9.57 \cos(200t + 28.3^\circ) \text{ A}$$

14) The average power supplied by the source is:

$$P_s = \frac{1}{2} V_m I_m \cos(\theta_v - \theta_i) = \frac{1}{2} (100)(9.57) \cos(45^\circ - 28.3^\circ) = 458.318 \text{ W}$$

15) The average power absorbed by the resistor is:

$$P_R = (10)(9.57 \angle 28.3^\circ) = 95.7 \angle 28.3^\circ \text{ W}$$

16) The average power absorbed by the inductor and the capacitor is:

$$P_L = 0 \text{ W} \text{ and } P_C = 0 \text{ W}$$

Prob. 4: Consider the circuit shown in Figure 3 and answer questions 17-18.

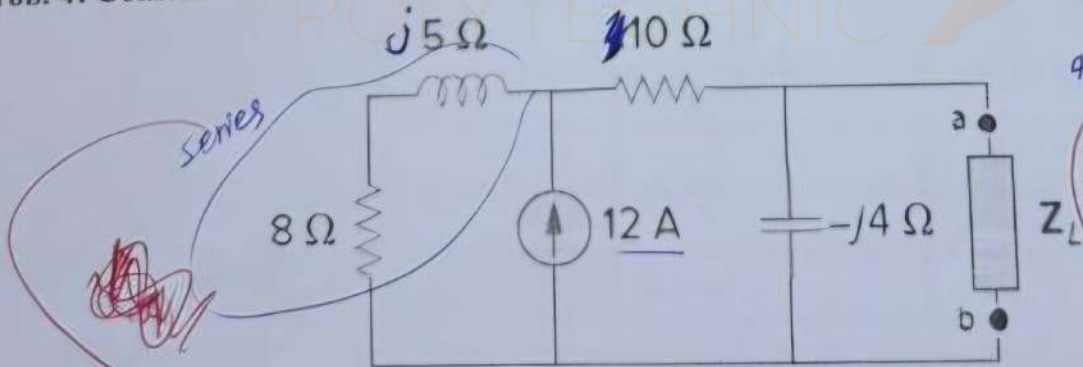


Figure 3

17. The Thevenin impedance with respect to terminals a-b is:

$$Z_{th} = (8 + j5 + 10) \parallel (-j4) = 8.86 - j4.04 \Omega$$

18. The Thevenin voltage with respect to terminals a-b is:

$$V_{th} = (113.2 \angle 32^\circ) \frac{-j4}{8 + j5 + 10 - j4} = 25.11 \angle -61.17^\circ \text{ V}$$

